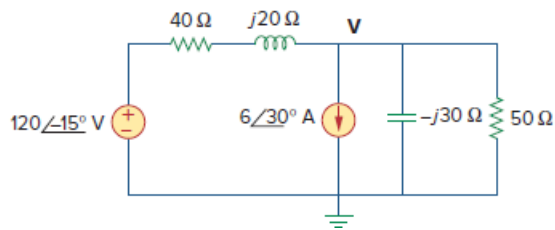


Problem

Use nodal analysis to find $\mathbf{V}$ in the circuit of Fig. 10.56.

Figure 10.56:



Step-by-step solution

Step 1 of 2

Refer to the circuit diagram depicted in Figure 10.56 in the textbook.

Apply Kirchhoff's current law at node $\mathbf{V}$.

$$\begin{aligned} \frac{\mathbf{V} - 120\angle -15^\circ}{40 + j20} + 6\angle 30^\circ + \frac{\mathbf{V}}{-j30} + \frac{\mathbf{V}}{50} &= 0 \\ \frac{\mathbf{V}}{40 + j20} - \frac{120\angle -15^\circ}{40 + j20} + 6\angle 30^\circ + \frac{\mathbf{V}}{-j30} + \frac{\mathbf{V}}{50} &= 0 \\ \mathbf{V} \left(\frac{1}{40 + j20} + \frac{1}{-j30} + \frac{1}{50} \right) &= \frac{120\angle -15^\circ}{40 + j20} - 6\angle 30^\circ \\ \mathbf{V} (0.02 - j0.01 + j0.033 + 0.02) &= 2 - j1.7803 - 5.196 - j3 \end{aligned}$$

Step 2 of 2

Simplify the equation further,

$$\begin{aligned} \mathbf{V} (0.04 + j0.0233) &= -3.196 - j4.78 \\ \mathbf{V} (0.0463\angle 30.22^\circ) &= 5.75\angle -123.76^\circ \\ \mathbf{V} &= \frac{5.75\angle -123.76^\circ}{0.0463\angle 30.22^\circ} \\ &= 124.19\angle -153.98^\circ \text{ V} \end{aligned}$$

Therefore, the value of voltage, $\mathbf{V}$ in the circuit is $\boxed{124.19\angle -153.98^\circ \text{ V}}$.